

Towards an Ontology of Memes

A Cultural Analytics Study on Viral Images

CANDIDATE

Alice Picco

SUPERVISOR

Fabio Vitali

CO-SUPERVISORS

Tommaso Campagna · Carlo Teo
Pedretti

SESSION

Session I · A.Y. 2025–2026

A Gap — and Three Lenses to Close It

Internet memes are among the most significant — and least formally organized — bodies of visual culture of the twenty-first century.

No domain ontology grounded in bibliographic and iconological theory has classified meme templates, their visual properties and their variant relationships *at corpus level*.

CLOSEST RELATED WORK

IMKG — Tommasini et al., 2023

A knowledge graph built for entity enrichment and downstream tasks.

THREE THEORETICAL LENSES

01

FRBR / LRM

The bibliographic hierarchy — Work, Expression, Manifestation, Item — separates abstract concept, named template and specific variant.

02

Panofsky's Iconology

Three semantic levels — pre-iconographical, iconographical, iconological — describe what an image shows and what it means.

03

Cultural Analytics

Distant viewing of large visual corpora surfaces patterns invisible to close reading — integrated with archival theory for web content.

Four Research Questions

RQ1

ONTOLOGICAL

What formal framework can classify memes as cultural objects — and how must FRBR and Panofsky be extended and integrated to fit them?

RQ2

COMPUTATIONAL

To what extent can CLIP zero-shot classification operationalize the pre-iconographical level across a significant dataset of images — and what are its limits?

RQ3

CULTURAL
ANALYTICS

What temporal, relational and platform-specific patterns become visible through corpus-level analysis of meme culture?

RQ4

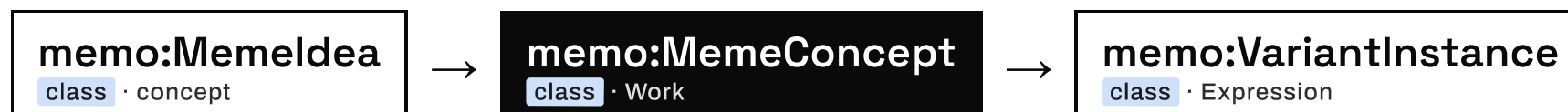
ARCHIVAL

How can community archives like KnowYourMeme become structured corpora while preserving the integrity of their curated metadata?

A Two-Axis Classification Model

Both axes are **OWL annotation tags** on each class (`frbrLevel · semanticLevel`) — the arrows show progression, not subclass or individual relations.

AXIS 1 — BIBLIOGRAPHIC (FRBR / LRM)



AXIS 2 — SEMANTIC (PANOFSKY)



THE EXTENSION — TRANSFORMATION TYPOLOGY

memo:TransformationDimension × memo:TransformationExtent model how variants depart from a template — a contribution to bibliographic modeling with no equivalent in FRBR.

FRBR / LRM in Practice: Distracted Boyfriend

THE EXAMPLE



COLOR KEY

- class
- individual
- object property
- data property

Conceptual Entity

The abstract idea the template crystallizes — redirected desire — independent of any image.

memo:MemeIdea class

memo:conceptDescription data property

Work

The named, community-documented template every version realizes.

memo:MemeConcept class

wdp:P123 (publisher → platform of origin) object property

memo:views data property

Expression

A specific labeled remix.

memo:VariantInstance class

wdp:P361 (part of → isVariantOf) object property

Manifestation

That variant as a file in a given encoding.

dcterms:format object property

Item

One localized copy — its path, CDN URL and scrape timestamp.

memo:imageFilePath data property

memo:variantImageURL data property

dcterms:modified data property

One Work, Many Expressions

`memo:MemeConcept (Work) – wdp:P527 (has part → hasVariant) → memo:VariantInstance (Expression)`

WORK `memo:MemeConcept` `class`



The named template — the blank canvas every version realizes.

EXPRESSION 1 `memo:VariantInstance`

`class`



A labeled variant — one Expression of the Work.

EXPRESSION 2 `memo:VariantInstance`

`class`



Another (differently) labeled variant — a distinct Expression.

Panofsky in Practice: Distracted Boyfriend

THE EXAMPLE



COLOR KEY

class individual object property data property

Pre-iconographical

CLIP

A colour photograph of three people; a man turns from one woman toward another.

Pure perception — no cultural knowledge required.

memo:hasImageType = memo:Photograph object property → individual

memo:hasSubjectMatter = memo:MultipleSubjects object property → individual

memo:isColor = true data property

Iconographical

COMMUNITY

Recognized as an **Exploitable** only by reading the **labels applied to each of the three figures** — an identification that requires shared cultural conventions.

memo:Exploitable individual memo:ImageManipulationFormat class

wdp:P2283 (uses) object property

Iconological

MANUAL

A metaphor of **redirected desire** — temptation, fickleness, the lure of the new over the committed. Hermeneutic judgement of meaning.

memo:hasReference → memo:SocialPhenomenon object property → class

memo:CulturalReference class

From Community Archive to Research Corpus

STAGE 01 Scrape 5,000 entries from KnowYourMeme by random sampling — drawn from 21,390 confirmed entries — with five structured metadata fields.	STAGE 02 Model OWL ontology in Protégé, structured along the two classificatory axes.	STAGE 03 Populate CLIP ViT-B/32 zero-shot on 5,000 images, plus programmatic metadata assertion and manual D0 annotation.	STAGE 04 Visualize Five interactive D3.js visualizations in the MEMO web application.
--	---	---	---

DATASETS — THREE NESTED SAMPLES

GLOBAL CORPUS 5,000 entries Random sample scraped from KnowYourMeme — drawn from 21,390 confirmed entries — characterizes the general population of meme culture documented on KYM.	D0 SAMPLE 50 entries Subset drawn from the corpus, chosen partially manually, partially randomly — basis for full manual annotation.	VARIANT DATASET 391 variants from 43 D0 entries Variant images scraped from the 43 of 50 D0 entries with galleries (≤10 each) — populates the Expression level.
---	--	---

The Complete MEMO Vocabulary

Class / ▶ subclass **individual** controlled-vocabulary value

BIBLIOGRAPHIC — FRBR / LRM

MemeIdea **MemeConcept**

VariantInstance

ICONOLOGICAL

CulturalReference

▶ **WebCulture**

▶ **PublicFigure**

▶ **HistoricalEvent**

▶ **SocialPhenomenon**

TRANSFORMATION

TransformationDimension

CaptionChange · VisualSubstrate · MediumShift · StyleShift · CompositionShift · CrossoverMerge · LanguageShift · Localization

TransformationExtent

Minimal · Moderate · Substantial · Parody

FORMAT

MemeFormat ▼ 6 subclasses

▶ **VideoFormat**

ViralVideo · ViralDebate · LipDub

▶ **TextFormat**

Catchphrase · Copy pasta · Slang · Snowclone · Hashtag

▶ **ImageManipulationFormat**

Exploitable · ImageMacro · Photoshop · Remix

▶ **ParticipatoryFormat**

ParticipatoryMedia · Dance · SocialGame

▶ **NarrativeFormat**

Character · FanArt · Parody · PopCultureReference · Reaction

▶ **AudioFormat**

Song · SoundEffect

CONTROLLED VOCABULARY

ImageType

Photograph · Cartoon · Drawing · Illustration · Painting

SubjectMatter

PersonPresent · CharacterPresent · AnimalPresent · MultipleSubjects · ObjectOnly · TextOnly · Unknown

FileFormat

JPEG · PNG · GIF · WebP · BMP

AnimationStatus

Static · Animated

TimePeriod **Region**

temporal & geographic vocab

OBJECT PROPERTIES · 28

MEMO: — BY PRIMARY DIRECTION
 isConceptualizedAs · hasReference · hasImageType · hasSubjectMatter · hasAnimationStatus · hasTransformationDimension · hasTransformationExtent · isFormatOf · hasOriginWork · isOriginPlatformOf · isRegionOf · isTimePeriodOf

REUSED — WIKIDATA / DUBLIN CORE

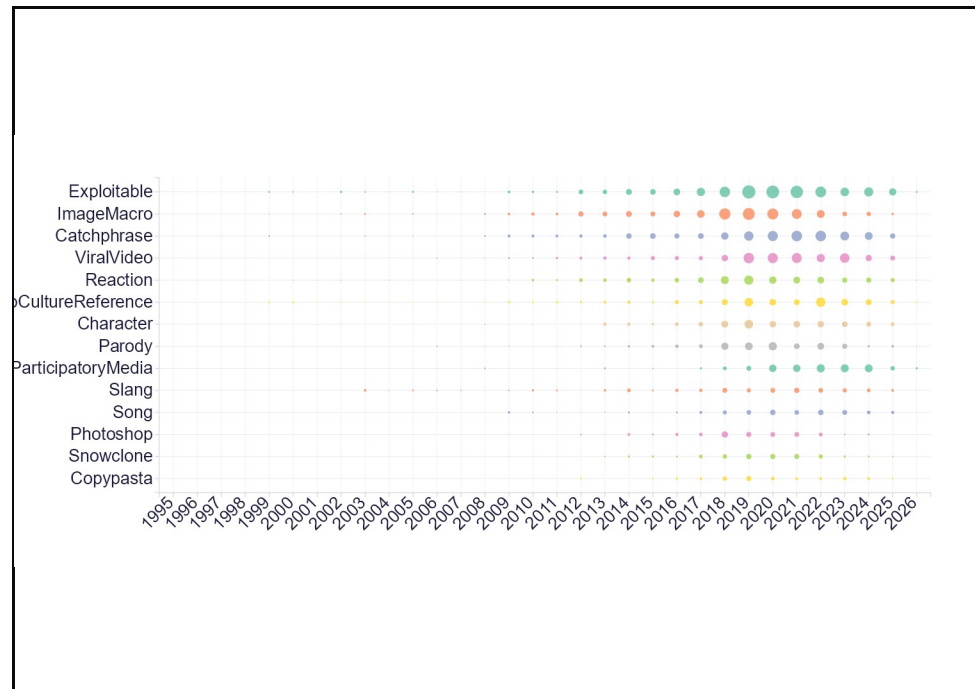
wdp:P123 (platform of origin) · **wdp:P2283** (uses) · **wdp:P571** (inception) · **wdp:P495** (country of origin) · **wdp:P361** (part of → isVariantOf) · **wdp:P527** (has part → hasVariant) · **dcterms:format**
13 inverse pairs (26) + dcterms:format + prov:wasDerivedFrom (no inverse) = 28.

DATA PROPERTIES · 14

MEMO:
 isColor · hasTextPresence · conceptDescription · views · hasId · imageFilePath · imageFilename · variantImageURL
EXTERNAL VOCABULARIES
 dcterms:created · dcterms:description · dcterms:modified · schema:url · schema:image · schema:keywords

How the Dominant Formats Spread Over Time

FORMAT TYPE × TEMPORAL DISTRIBUTION



READING THE FORMAT TYPES (the 14 charted)

Exploitable — a blank template made to be relabeled

Distracted Boyfriend

ImageMacro — an image with caption text

Catchphrase — a repeated, memorable phrase

ViralVideo — a clip that spreads and is reused

Reaction — an image used to express a reaction

PopCultureReference — a nod to film, TV or music

Character — a recurring persona or figure

Parody — a comic imitation of a work

ParticipatoryMedia — a trend or challenge people join

Slang — a coined word or expression

Song — a musical clip or track

Photoshop — a digitally composited image

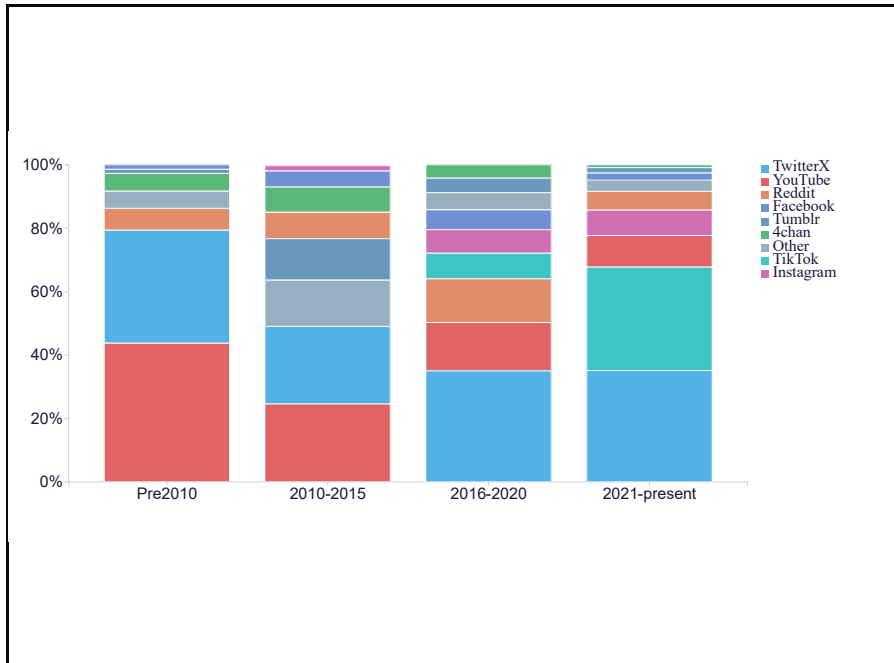
Snowclone — a fill-in-the-blank phrase template

Copypasta — a text block copied verbatim

Exploitable & Catchphrase are foundational and durable across the timeline; TikTok-native participatory formats cluster post-2021.

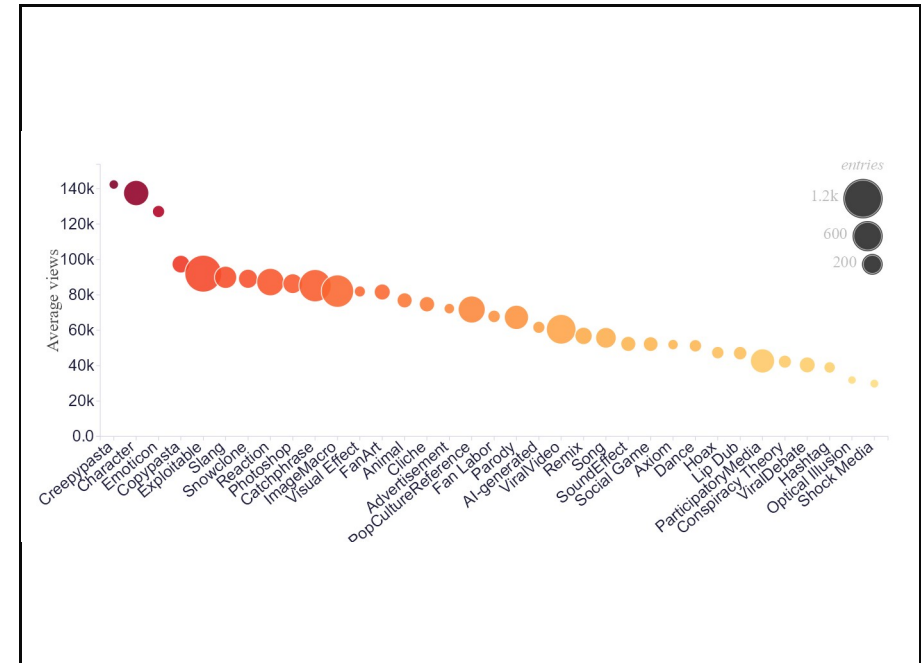
Platform Succession & Reach

PLATFORM OF ORIGIN × PERIOD



YouTube & Twitter (pre-2010) → plurality → Twitter/X dominance
→ TikTok & Twitter/X (2021–present).

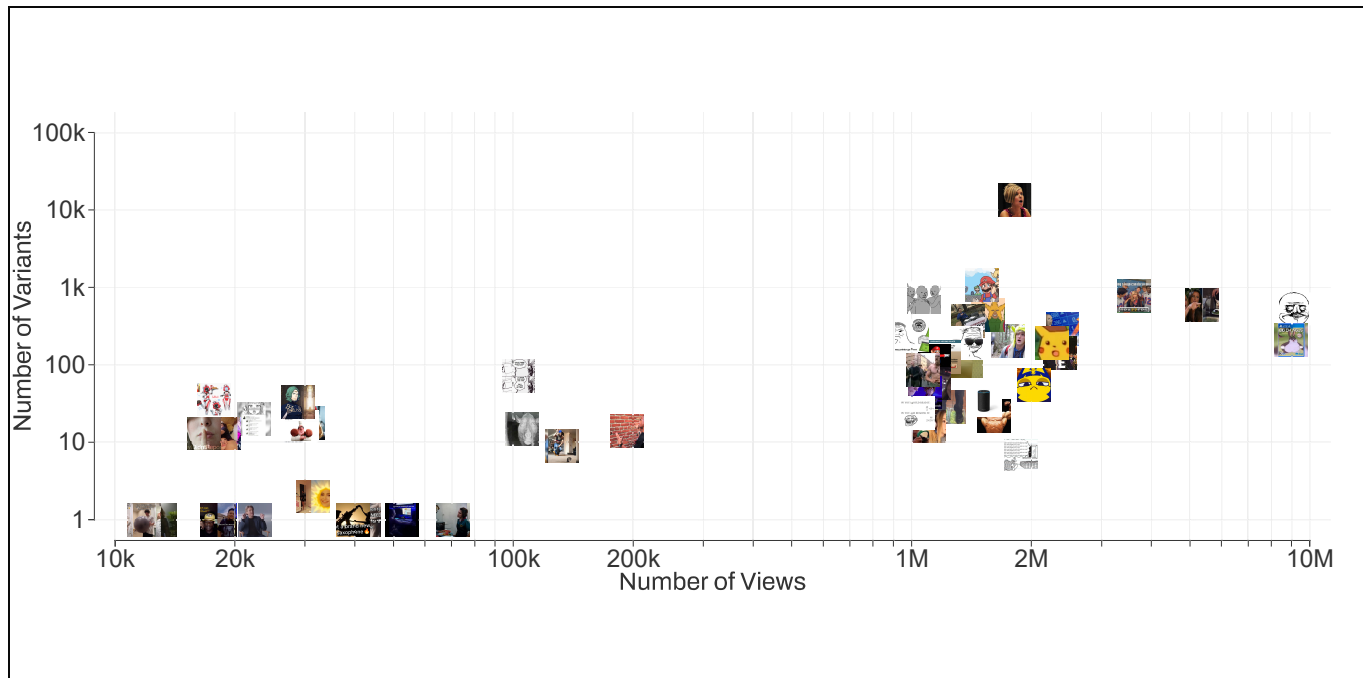
AVERAGE VIEWS BY FORMAT



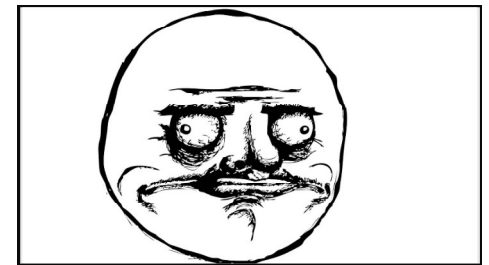
Format predicts reach: Creepypasta, Character & Emoticon
formats draw disproportionate views relative to corpus share.

Productivity $\neq$ Prominence

D0 — VARIANT COUNT $\times$ VIEW COUNT



Karen — 13,000+ variants · 2M views



Me Gusta — hundreds of variants · 9M views

Karen — 13,000+ variants at 2M views · **Me Gusta** — hundreds of variants at 9M views. The decoupling is visible only once Work (views) and Expression (variants) levels are distinguished.

Variants by Transformation Extent

A SUBSTANTIAL TRANSFORMATION, NEXT TO ITS ORIGINAL

Original



Variant (Expression)

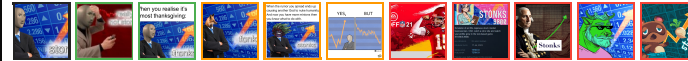


Substantial transformation: a pumpkin painted to resemble the canonical template.

THE ANNOTATED SET — BY TRANSFORMATION EXTENT

Transformation extent: ● Minimal ● Moderate ● Substantial ● Parody

Stonks



Me And The Boys



Npc Wojak



Surprised Pikachu



Steven Crowders Change My Mind Campus Sign



Variants range from **minimal** caption swaps to **substantial** re-compositions — information the Expression level carries on its own.

Answering the Research Questions

RQ1 — Ontological ✓

FRBR + Panofsky + the transformation typology classify memes without flattening their cultural complexity. The typology is a genuine contribution to bibliographic modeling.

RQ3 — Cultural Analytics ✓

Corpus-level analysis surfaces format hierarchies, temporal dynamics and platform dependencies invisible to close reading — meme culture revealed as a structured system.

RQ2 — Computational ❶ partial

CLIP *partially* operationalizes the pre-iconographical level but produces systematic hallucination. ImageType is plausible (Cartoon 45.3% / Photograph 44.8%); SubjectMatter is 34.4% Unknown. Completeness was traded for reliability.

RQ4 — Archival ✓

A community archive can become a structured corpus when its descriptors are kept honest about their origin — e.g. wdp:P123 (publisher → platform of origin) is stored as a community *attribution*, not an objective fact. Shown for the D0 sample.

Conclusions & Future Developments

MEMO is a working OWL 2 DL ontology of internet memes, built and populated from a 5,000-entry KnowYourMeme corpus (sampled from 21,390 confirmed entries).

It shows that community-curated archives can be turned into structured research corpora *without discarding* the cultural information in their metadata — as long as the transformation stays transparent about the interpretive character of the source vocabulary.

Its deeper semantic and variant levels are demonstrated on a 50-entry sample (D0); extending them across the full corpus, and validating the automated steps, remain open work.

FUTURE DEVELOPMENTS

- Populate the FRBR conceptual level across the full 5,000-entry corpus (LLM-assisted, human-validated).
- Complete the Expression level: annotate all 391 variants and extend variant scraping.
- Extend the memo:CulturalReference layer beyond the D0 dataset.
- Model explicit Expression individuals and LRMoo creation events (F27 / F28).
- Publish the controlled vocabularies as SKOS.